

Identification of water flow through non-root soil macropores and along roots in shrub-encroached grassland

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Abstract

Identification of water flow through non-root soil macropores and along roots using X-ray computed tomography (CT) is still challenging because of limited methods to quantify roots and pore structures. This study selected shrub-encroached grassland consisting of a mosaic of shrub patches and interspace grass patches and the original grassland as a control treatment. Twelve intact soil columns (0–30 cm deep) were excavated, which included four replicates each from the soils under the original grassland, the shrub patches and interspace grass patches. Three-dimensional networks of soil macropores and roots were quantified using CT scanning and image analysis techniques, and active flow pathways were identified through KI tracing experiment using CT scanning. The results showed that the tracer depths of soils under the shrub patches and interspace grass patches were deeper than those of the original grassland, which was attributed to the well-developed root system. The tracer volume density was significantly and positively correlated with soil macropore node density and root volume density ($p < 0.05$), and significantly and negatively correlated with root mean angle ($p < 0.05$). Soils under the shrub patches had lower tracer volume density than that under the interspace grass patches, which was attributed to the horizontal-oriented roots under the shrub patches. The relative contribution of water flow along roots was larger than that through non-root soil macropores 35 min after dye application. This provided direct evidence that shrub roots were the main pathway of macropore flow resulting in water flow into deep soil layers in the shrub-encroached grassland.

Highlights

- The tracer depths of shrub-encroached grassland were deep due to well-developed root system.
- Shrub patches had low tracer volume density due to the horizontal-oriented roots.
- The contribution of water flow along roots was larger than that through non-root soil macropores.
- Shrub roots were the main pathway of water flow.

KEYWORDS

grassland, root, shrub encroachment, water flow, X-ray CT

1 | INTRODUCTION

Soil water flow is an important hydrological process that links surface water and groundwater (Wang et al., 2008; Zhang et al., 2019). Studies have shown that preferential flow could increase the leaching risk of soil, resulting in accelerated movement, of varying extents, of nutrients (Stamm et al., 1998), pesticides (Jørgensen et al., 1998) and pathogens (Darnault et al., 2004). There are many factors including soil properties (bulk density, soil organic carbon, soil texture and soil aggregates), irrigation technique, tillage method and biological factors, that affect soil water infiltration (Koestel et al., 2012; Koestel & Larsbo, 2014; Lipiec et al., 2006). Although soil macropores only contribute a small portion to total soil porosity, they play a pivotal role in generating water flow (Moret & Arrué, 2007). Among these factors, plant roots significantly affect soil water infiltration in arid areas (Devitt & Smith, 2002; Wu et al., 2017). Roots may create channels to promote water flow because there is an air gap between the growing root and the soil, which are important conduits for downward water movement (Nespoulous et al., 2019; Van Schaik, 2009). Germann et al. (2012) indicated that root channels increase macropore network density and continuity, and channels formed by plant roots may be conducive to fast water and solute movement. Wu et al. (2017) found that the initial and average infiltration rate during the first 15 min increased with the increasing diameter of root channels. However, identification of the type of macropores initiating water flow is still complicated.

X-ray computed tomography (CT) is a non-destructive imaging method to study soil structure in contrast to the traditional methods, such as dye tracing and soil thin-sectioning (Capowiez et al., 2003; Grevers et al., 1989). Over the past decade, CT has been employed to quantify the characteristics of soil macropores (Katuwal et al., 2015; Luo et al., 2010). Recently, some studies have been conducted to quantify roots using CT. Koebernick et al. (2018) investigated the dynamics of root-soil contact during soil drying using CT. Kuka et al. (2013) visualised and quantified soil structure and roots simultaneously using X-ray CT. Hu et al. (2019) quantified the root architecture of soil cores using X-ray CT. In addition, CT has become an attractive technology to non-destructively observe solute transport and preferential flow in soil (Luo et al., 2008; Soto-Gómez et al., 2018). Soto-Gómez et al. (2018) linked soil pore characteristics to water flow and tracers in soils using an

industrial CT scanner. The above-mentioned studies indicate that CT is an appropriate technique for quantifying preferential flow, soil macropores and roots in undisturbed soils (Mees, 2003; Kalender, 2005; Helliwell et al., 2013; Sammartino et al., 2015). However, the visualisation and quantification of the real-time water flow using X-ray CT method remains challenging.

Shrub encroachment is a global environmental phenomenon, which is widely distributed in arid, semiarid and subalpine regions (Parizek et al., 2002; Van Auken, 2009). Data show that more than 5.1×10^6 ha of grassland have been invaded by *Caragana microphylla* Lam in the Inner Mongolia grassland of northern China (Li, Zhang, et al., 2013). Some studies have reported that shrub encroachment causes increases the aboveground biomass (Hancock et al., 2008; Iponga et al., 2009), nutrient cycling changes (Barton et al., 2005) and changes in underground biological processes (Marschner et al., 2004). Shrub encroachment has a great effect on soil structure and soil water distribution at scales of individual plants, patches, slopes and watersheds (Hu et al., 2015; Li et al., 2009). Li, Hu, et al. (2013) reported that rainfall partitioning by shrub canopies and subsurface soil macropores induced by root architecture resulted in preferential water flow into deeper soil layers, which might favour competitive advantages for water by shrubs under arid conditions. However, there was no direct evidence to prove that pores formed through shrub roots were the main pathway for water flow into deep soil layers. Therefore, exploration of methods and technologies to identify water flow through pores formed by shrub roots is urgently needed. To fill the above knowledge gaps, we hypothesize that water flow varies significantly between the shrub-identified grassland and the original grassland, and water flow induced by shrub roots transports water into deep soil layers. The specific objectives were to: (1) visualise and quantify networks of water flow, soil macropores and roots using CT and dye tracer; (2) confirm that shrub roots induce water flow into the deep soil following shrub encroachment.

2 | MATERIALS AND METHODS

2.1 | Study sites and soil column sampling

The soil sampling sites were located in Taibus Banner, Inner Mongolia in northern China. The study area has a

FIGURE 1 Landscape of the field sites for the original grassland, shrub patch and interspace grass patch



temperate semiarid climate. The mean annual temperature, precipitation and potential evapotranspiration are 1.6°C, 407 mm, and 1750–2150 mm, respectively. The soils are chestnut soils equivalent to Calcic-orthic Aridisols (Soil Survey Staff, 1987). The vegetation consists of *Caragana microphylla* (shrub), *Stipa krylovii* (grass), *Leymus chinensis* (grass) and *Artemisia frigida* (grass).

Three experimental plots were selected (20 m × 20 m): the original grassland (the control treatment), shrub patches and interspace grass patches in shrub-encroached grassland. The shrub size was $4.34 \pm 1.08 \text{ m}^2$ and the shrub cover was 40.1% in the shrub-encroached grassland (Figure 1). Intact soil columns were collected at intervals of 5–10 m in each plot, and soil samples for analysing physico-chemical properties were collected by digging profiles of 30 cm × 30 cm × 30 cm (width × height × depth) after the excavation of the soil columns. The aboveground shrub or grass was carefully clipped to ground level with minimum disturbance, and litter was completely removed to expose bare soil before soil sampling. We moistened the soil with water before sampling and the core was taken with polyvinyl chloride (PVC) cylinder (10 cm external diameter and 30 cm length) with a bevelled edge at its bottom. Core extraction followed the procedure of Sammartino et al. (2012). The core was extracted by pushing, slowly and vertically, into the soil, the penetration of the cylinder progressing centimetre by centimetre, carefully removing the soil around the tube bottom at each step, to minimise mechanical stresses during extraction. After sampling, a PVC grid was glued to the base of the cylinder, and the cylinder was then glued to an annular sample holder to protect the bottom of the core from mechanical disturbance. To protect the samples during transport from field to laboratory, the cores were placed in sealed plastic bags and were wrapped with a sponge and packed with wheat straw. Four replicates were sampled from the original grassland, the shrub patches and interspace grass patches, respectively. Therefore, there were 12 intact soil columns.

Soil samples near the soil columns were collected at 0–30 cm to determine soil properties. Sand fraction

(0.05–2 mm) was measured by sieving, and the silt (0.002–0.05 mm) and clay (<0.002 mm) fractions were determined using the pipette method (Klute, 1986). The soil texture was then determined according to the particle size distribution. The soil samples were oven-dried at 105°C for 12 h to determine soil bulk density. Soil organic carbon content was determined by the method of Walkley and Black (1934) and converted to organic matter content by multiplying by 1.724. The soil properties at a soil depth of 0–30-cm are listed in Table 1.

2.2 | Soil macropore and root

2.2.1 | CT scanning

A GE industrial scanner (phoenix nanotom[®] m) was used to scan the soil columns at an energy level of 210 kV and 235 μA, which required 35 min to scan one column each time. Soil columns were scanned twice: the first scanning was for identifying and quantifying soil macropore and root structure, and the second scanning was for identifying paths of water flow with KI tracer. Scanning for soil macropores and roots took place before saturating the columns with deionised water. The number of projections and the exposure time was 1800 and 333 ms, respectively. The distance between the scanned sample and the source was 55 cm. After reconstruction, 1500 slices with 1700 × 1700 pixels per slice for each column were produced, leading to the voxel size of 143 μm × 143 μm × 143 μm.

2.2.2 | Soil macropore segmentation and analysis

The interpretation of soil macropores followed the procedure of Luo et al. (2010) in Avizo 9.0 (FEI, 2016). A cylindrical cropping tool was applied to obtain a region of interest (ROI), and then the area outside the ROI was deleted to

TABLE 1 Soil properties of the original grassland and shrub-encroached grassland (interspace grass patches and shrub patches)

Treatments	Soil depth (cm)	Organic matter (g kg ⁻¹)	Bulk density (g cm ⁻³)	Texture	Particle size composition (%)		
					0.05–2.00 mm	0.002–0.05 mm	Φ < 0.002 mm
Original grassland	0–30	15.17 (±1.32)	1.42 (±0.21)	Sandy loam	59.59 (±3.87)	24.33 (±1.54)	16.08 (±0.35)
Interspace grass patches	0–30	26.71 (±0.89)	1.51 (±0.17)	Sandy loam	59.12 (±5.34)	24.00 (±1.97)	16.88 (±0.98)
Shrub patches	0–30	31.88 (±0.98)	1.59 (±0.05)	Sandy loam	54.12 (±5.76)	29.67 (±3.24)	16.21 (±1.54)

exclude voids near the core walls and to remove the cracks that formed on the margin of soil columns during sampling disturbances. The threshold value of soil macropores (<12,000) was selected according to the grey scale histogram (Figure 3a). The selected threshold value was then applied to the all slices of ROI to binarize the data. The grey histogram was obtained for the whole column and the same uniform procedure was carried out for all columns. The soil macropore networks were visualised by volume rendering.

After segmentation, soil macroporosity, surface area density, number density, mean equivalent diameter, and the size distribution of macropores (categorised by diameter) were calculated (Hu et al., 2019). Skeletonization of macropores is important to accurately quantify the topology of soil macropores by determining the nearest distance to the border voxels. After skeletonization, macropore number density (ρ_d [m⁻³]), the node density (ρ_n [m⁻³]), the length density (g_l [km m⁻³]), the mean tortuosity (τ), the mean angle and the mean equivalent radius were all calculated (Hu et al., 2019).

After skeletonization, macropore number density (ρ_d , m⁻³) was defined as the number of macropore networks (N_m) in a unit volume (V):

$$\rho_d = \frac{N_m}{V} \quad (1)$$

A node refers to an intersection at which two macropore branches are connected. The number of nodes in each macropore network was recorded, and we calculated the node density (ρ_n , m⁻³) as the number of nodes (N_n) in a unit volume (V):

$$\rho_n = \frac{N_n}{V} \quad (2)$$

The length density (ρ_l , km m⁻³) was also analysed as the total actual macropore length (L_t) in a unit volume (V ; Luo et al., 2010).

$$\rho_l = \frac{L_t}{V} \quad (3)$$

The mean tortuosity (τ) was calculated as a ratio of the total actual macropore length (L_t) to the total straight-line distance (L_l) of all the macropores in a certain volume (Luo et al., 2010). The mean tortuosity means the ratio of total pore length to that of the shortest possible path.

$$\bar{\tau} = \frac{\sum_{i=1}^n L_{ti}}{\sum_{i=1}^n L_{li}} \quad (4)$$

where i is the index of a macropore branch and n is the total number of macropore branches.

The inclination of a macropore branch was characterised by an angle (θ , °) away from a vertical direction. The mean angle was calculated as:

$$\bar{\theta} = \frac{\sum_{i=1}^n L_{ti}\theta_i}{\sum_{i=1}^n L_{li}} \quad (5)$$

Assuming that all macropores are cylindrical, its mean equivalent hydraulic radius (mm) was calculated by the total volume V_t and total length L_t of the macropore:

$$\bar{r} = \sqrt{\frac{V_t}{L_t\pi}} \quad (6)$$

2.2.3 | Root segmentation, analysis and validation

Roots were quantified through a combination of the growth method and the threshold method (Hu et al., 2018; Kuka et al., 2013). Some impurities that were not roots might also be selected if only the threshold value was used. Therefore, a coarse range of threshold values (12000–15,000) was first distinguished according to the grey value histogram (Figure 2a). The magic wand

tool was then used to select roots in the 2-D images, and the root morphology was displayed in the window after applying to all slices. It was necessary to enlarge the selected area to ensure that the inner valid voxels were counted as roots. The root networks were then visualised and quantified. The root volume density, surface area density, length density, mean equivalent diameter, mean root volume and mean angle was calculated. Root volume density, surface area density and length density were defined as the ratio of the root volume, total surface area and total length to the volume of ROI. The inclination of a root branch was characterised by an angle (θ) away from the vertical direction, and the mean angle of the root was calculated in the same way as for pores (Equation 5). The effectiveness of this method for root extraction has been shown by Hu et al. (2019). The correlations between the estimated root length and root volume determined by X-ray CT and those determined by the standard root washing method are shown in Figure 4. There were good correlations between the root lengths, with a calculated coefficient of determination of $R^2 = 0.97$, and root volumes, with $R^2 = 0.92$, determined by the two methods (Figure 4). These results show that X-ray CT is a suitable method for root evaluation in situ. After scanning, the 3 cores were cut into three sections (0–100 mm, 100–200 mm and 200–300 mm) and the roots were collected. The roots of 9 sections were subjected to a manual washing procedure with a 0.25-mm sieve to separate the individual roots from the soil (Kuka et al., 2013). Root volume and length were quantified by the method of a standard root washing with the software WinRhizo 2009 (regent instruments: Arsenault et al., 1995).

2.3 | Water flow

2.3.1 | KI dye tracing experiment

Soil columns were wrapped with gauze at the bottom. They were then placed into deionised water to soak for 24 h for saturation, removed and allowed to drain for another 24 h to remove gravity water. The purpose of this was to allow the tracer to flow mainly through the macropores in the soil.

Due to the high X-ray attenuation value of KI solution and its capacity for increasing contrast (Luo et al., 2008; Wildenschild et al., 2002), we used KI solution to trace water flow. Soil with and without KI solution ($0.5 \text{ g}\cdot\text{ml}^{-1}$) was added into the soil column (10 cm external diameter and 3.2 cm length) for a pre-tracer experiment. The soil without KI solution was placed in zone A, and the soil with KI solution was placed in zone B (Figure 3). The results show that the

KI solution is reliable as a tracer to study water flow and that the threshold of soil with KI solution is about $>28,000$ (grey value).

Soil columns were placed in the GE industrial scanner (phoenix nanotom[®] m) during the KI dye tracing experiment to minimise disturbance by moving soil columns during the scan. First, 100 ml KI solution with a concentration of $0.5 \text{ g}\cdot\text{ml}^{-1}$ (i.e., 15 mm) was applied to each column on the surface. At the same time, the scanning was carried out at the beginning of the dye tracer experiment. The scanning was completed after 35 min. These images were used to interpret the water flow for 35 min. After the first scanning, another 250 ml KI solution with a concentration of $0.5 \text{ g}\cdot\text{ml}^{-1}$ (i.e., 37.6 mm) was immediately applied to each column on the surface. The scanning time was also 35 min, therefore, the total tracing time was 70 min. These images were used to interpret the water flow for 70 min. During these two scans, the tracer did not flow to the bottom of the columns. The selection of tracer content and scanning time was determined in preliminary experiments and was chosen so that scanning accuracy was as high as possible and no tracer was lost at the bottom of the columns.

2.3.2 | Water flow segmentation and analysis

Slices scanned from the KI dye tracer experiment were used to interpret water flow. A cylindrical cropping tool was applied to obtain a region of interest (ROI) to remove obvious disturbances. The threshold value of KI was selected according to the pre-tracer experiment and was adjusted slightly according to the image. The selected threshold value of KI (26000) had no attenuation before applying KI (Figure 2a), which confirmed the reasonability of the threshold. After applying the selected threshold value of water flow to all slices, the water flow networks were visualised by volume rendering. To investigate the variations in tracer volume density with soil depth, the tracer volume density of each slice was calculated. The total tracer volume and tracer volume density (i.e., the tracer volume in a unit volume) was then calculated. The tracer volume density (35 min) means the tracer volume in a unit volume (volume of soil column) after 35 min of tracer infiltration.

2.4 | Statistical analyses

All statistical analyses except redundancy analysis (RDA) were conducted with IBM SPSS 20 (SPSS Inc., USA). RDA was conducted with the vegan package in R version 3.5.2 (R Core Team, 2021) with vegan package (Oksanen et al., 2020) to determine the relationship between water

flow and soil macropores and roots. The relative contribution was calculated as the absolute value of standardised regression coefficient/the sum of the absolute value of all standardised regression coefficients.

Homogeneity of variances and normal distribution of the data was tested before applying the parametric methods. One-way analysis of variance followed by the least significant difference method was used to compare water flow, soil macropores and roots between different treatments. Pearson correlations were conducted to evaluate the linkage among water flow, soil macropores, and roots in the different treatments (the original grassland, the shrub patches and interspace grass patches).

3 | RESULTS

3.1 | Soil macropore networks

3.1.1 | Visualisation and CT-measured soil macroporosity

Figure 5a shows three-dimensional visualisations of soil macropores of all soil columns, and Figure 6a shows changes in the soil macroporosity along the soil layer of all soil columns. The networks were deeper distributed under the shrub patches and interspace grass patches than that under the original grassland. More specifically,

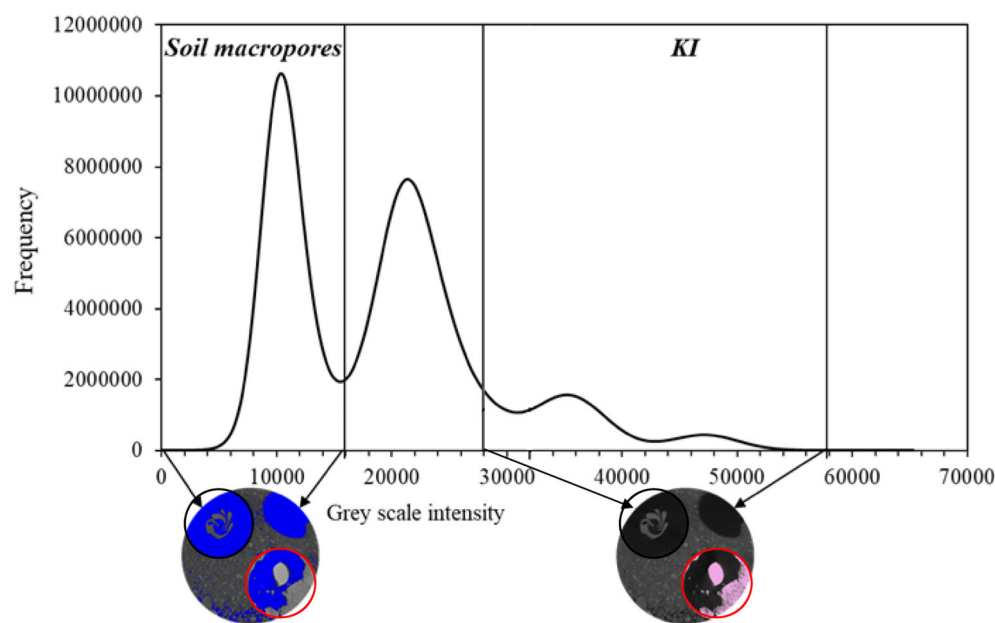


FIGURE 2 Grey values of soil columns scanned using CT after applying KI (preliminary experiment). Black circles (zone A) place soil without KI solution; red circles (zone B) place soil with KI solution; blue denotes soil macropores and pink denotes KI.

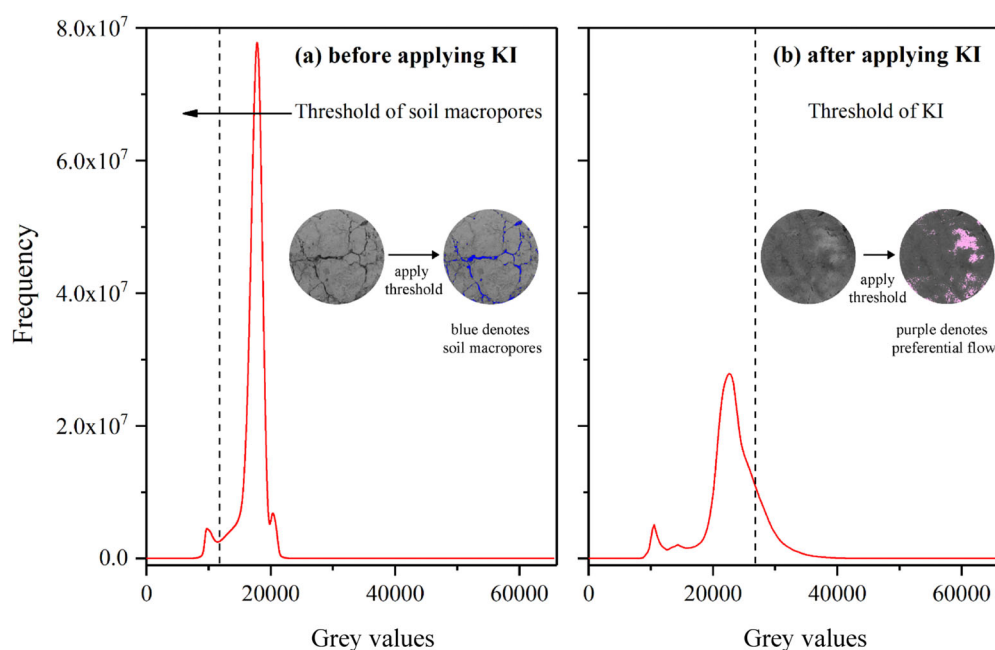


FIGURE 3 Grey values of soil columns scanned using CT before (a) and after (b) applying KI solution. (a) Was for the interpretation of soil macropores and roots, and blue denotes soil macropores; (b) was for the interpretation of water flow, and pink denotes KI.

TABLE 2 Distribution of pore diameter of soil macroporosity under original grassland, interspace grass patches and shrub patches ($\times 10^{-5} \text{ mm}^3 \text{ mm}^{-3}$)

Treatments	Pore diameter						
	0.1-1 mm	1-2 mm	2-4 mm	4-6 mm	6-8 mm	8-10 mm	>10 mm
Original grassland	18.24 (±5.23) b	21.64 (±2.91) b	32.17 (±12.91) b	20.63 (±6.96) b	51.00 (±32.87) b	12.84 (±22.25)	0
Interspace grass patches	22.13 (±6.18) b	30.29 (±5.81) b	45.55 (±6.77) ab	37.59 (±11.75) ab	38.87 (±28.9) ab	37.32 (±38.89)	89.44 (±94.07)
Shrub patches	40.53 (±12.72) a	53.48 (±12.83) a	78.69 (±27.65) a	69.4 (±32.41) a	68.01 (±45.65) a	40.06 (±26.76)	269.23 (±177.62)

Note: Column marked by the same letter is not significantly different at $p = 0.05$ between the treatments, the absence of letters indicates no significant differences. Data in the parentheses are standard deviations.

TABLE 3 Soil macropore characteristics of soil columns as determined using the CT images

Treatments	Surface area					Tortuosity	
	Macroporosity $\times 10^{-3}$ ($\text{mm}^3 \text{ mm}^{-3}$)	Length density $\times 10^3$ ($\text{mm} \cdot \text{mm}^{-3}$)	Surface area density $\times 10^3$ ($\text{mm}^2 \cdot \text{mm}^{-3}$)	Mean equivalent diameter (μm)	Hydraulic radius (mm)	Node density $\times 10^6$ ($\text{no} \cdot \text{m}^{-3}$)	Number density $\times 10^5$ ($\text{no} \cdot \text{m}^{-3}$)
Original grassland	1.467 (± 0.0006) b	3.46 (± 0.77) b	9.21 (± 1.96) b	446.60 (± 37.09)	0.45 (± 0.06)	2.49 (± 0.46) b	5.84 (± 1.24)
Interspace grass patches	3.042 (± 0.0011) b	4.94 (± 1.62) b	18.47 (± 7.37) b	432.50 (± 126.06)	0.44 (± 0.13)	4.23 (± 1.48) ab	7.55 (± 2.42)
Shrub patches	5.909 (± 0.0021) a	8.29 (± 2.83) a	30.03 (± 10.94) a	458.98 (± 125.36)	0.49 (± 0.14)	7.26 (± 2.63) a	13.25 (± 4.58)

Note: Column marked by the same letter is not significantly different at $p = 0.05$ between the treatments, the absence of letters indicates no significant differences. Data in the parentheses are standard deviations.

TABLE 4 Root characteristics of soil columns as determined using the CT images

Treatments	Root volume density ($\text{mm}^3 \cdot \text{mm}^{-3}$)	Mean equivalent diameter (mm)	Mean root volume (mm^3)	Mean angle ($^\circ$)	Surface area density ($\text{mm}^2 \cdot \text{mm}^{-3}$)	Length density ($\times 10^{-3} \text{ mm} \cdot \text{mm}^{-3}$)
Original grassland	0.0002 (± 0.0001) b	3.166 (± 0.359) b	55.838 (± 3.885)	38.240 (± 3.257)	0.004 (± 0.005) b	0.232 (± 0.053)
Interspace grass patches	0.0010 (± 0.0006) b	5.026 (± 1.743) ab	114.454 (± 10.633)	28.545 (± 3.545)	0.006 (± 0.003) b	0.408 (± 0.226)
Shrub patches	0.0034 (± 0.0017) a	6.676 (± 1.590) a	453.113 (± 15.543)	54.932 (± 2.378)	0.021 (± 0.014) a	0.543 (± 0.266)

Note: Column marked by the same letter is not significantly different at $p = 0.05$ between the treatments, the absence of letters indicates no significant differences. Data in the parentheses are standard deviations.

shrub patches had more soil macropores at the deep soil depth than interspace grass patches. For the morphology of soil macropores, the soil macropores under the shrub patches and interspace grass patches were larger, longer and more continuous than those under the original grassland.

Table 2 shows the pore diameter distribution of macroporosity. For the shrub-encroached treatments, macropores with diameters >10 mm dominated in soils under shrub patches ($269.23 \times 10^{-5} \text{ mm}^3 \cdot \text{mm}^{-3}$) and interspace grass patches ($89.44 \times 10^{-5} \text{ mm}^3 \cdot \text{mm}^{-3}$). For the original grassland, more macropores consisted of diameters between 6 and 8 mm ($51 \times 10^{-5} \text{ mm}^3 \cdot \text{mm}^{-3}$). Significant differences were found in macropores of diameter <6 mm between the shrub patches and the original grassland ($p < 0.05$). No significant difference was found in macropores of >2 mm between the shrub patches and interspace grass patches.

3.1.2 | Macropore number density, surface area density, length density and node density

The soil macropores under the shrub patches had a greater surface area density, length density, node density and number density than those under the interspace grass patches and the original grassland (Table 3). There were no significant differences between treatments for mean diameter, hydraulic radius, node density, number density and tortuosity (Table 3).

3.2 | Root networks

Figure 5b shows three-dimensional visualisations of the roots of all soil columns. Figure 6b shows changes in the root volume density along all soil columns. The root networks were deeper for the interspace grass patches and shrub patches than for the original grassland. The roots were distributed in the deeper soil layer for the shrub patches than for the interspace grass patches. However, the roots under the shrub patches and interspace grass patches had distinctly different orientations.

Shrub patches had a greater root volume density, surface area density, length density and mean volume than the interspace grass patches and original grassland (Table 4). However, the roots under the shrub patches had a larger mean angle (54.932°) away from the Y-axis than under the original grassland (38.24°) and interspace grass patches (28.545°), indicating that the roots were predominantly lateral

TABLE 5 Distribution of root diameter of volume density under original grassland, interspace grass patches and shrub patches ($\times 10^{-4} \text{ mm}^3 \cdot \text{mm}^{-3}$)

Treatments	Root diameter				
	1–2 mm	2–3 mm	3–4 mm	4–5 mm	>5 mm
Original grassland	0.090 (± 0.004) a	0.131 (± 0.034)	0.933 (± 0.005)	0.734 (± 0.000)	8.881 (± 10.940)
Interspace grass patches	0.082 (± 0.012) a	0.211 (± 0.143)	0.584 (± 0.000)	1.480 (± 0.982)	5.239 (± 5.725)
Shrub patches	0.034 (± 0.000) b	0.178 (± 0.013)	0.932 (± 1.019)	1.980 (± 1.157)	46.300 (± 41.330)

Note: Column marked by the same letter is not significantly different at $p = 0.05$ between the treatments, the absence of letters indicates no significant differences. Data in the parentheses are standard deviations.

and horizontal oriented in the shrub patches, and longitudinally oriented in the interspace grass patches and original grassland.

Table 5 shows the root diameter distribution of volume density. For the shrub-encroached treatments, roots with diameters >5 mm dominated in soils under shrub patches ($46.30 \times 10^{-2} \text{ mm}^3 \cdot \text{mm}^{-3}$) and interspace grass patches ($5.23 \times 10^{-2} \text{ mm}^3 \cdot \text{mm}^{-3}$). Significant differences were found in roots of diameter 1–2 mm between the shrub patches and the interspace grass patches, the original grassland ($p < 0.05$).

3.3 | Water flow networks

3.3.1 | Visualisation of water flow networks

Figure 5c,d shows 3-D visualisations of water flow networks with tracer times of 35 and 70 min, respectively. The water flow paths were deeper and longer in the soils under the shrub patches and the interspace grass patches than that under the original grassland. Moreover, there were more water flow paths in the deep soil layer under the shrub-encroached grassland than in the original grassland. We also found that the interspace grass patches had deeper and longer water flow paths than the shrub patches.

3.3.2 | CT-measured tracer characteristic parameter

The tracer volume density characteristics with depth were distinctly different between the original grassland, interspace grass patches and shrub patches (Figure 6c,d). The curves show higher amplitude of tracer density for soils under the shrub patches and interspace grass patches than the original grassland. For soils under the shrub patches, the tracer volume density showed a rapid increase at 0–10 mm and then a decrease at 10–80 mm before becoming stable at 80–200 mm. For soils under

the interspace grass patches, a high concentration plateau over 0–30 mm was followed by a decrease to 80 mm. For soils under the original grassland, the tracer volume density curve showed a rapid decrease at 0–30 mm, and then it became stable at 30–200 mm.

The soils exhibited higher tracer volume densities at the tracer time of 70 min than that at the tracer time of 35 min (Figure 7). The soils under the original grassland showed 21.13%, 2.61%, 0.07% and 0.01% tracer volume densities at soil depths of 0–50 mm, 50–100 mm, 100–150 mm and 150–200 mm, respectively. The soils under the interspace grass patches exhibited 63.72%, 52.16%, 14.97% and 3.79% tracer volume densities with increasing soil depths. The tracer volume densities for soils under the shrub patches were 66.43% at 0–50 mm, 19.82% at 50–100 mm, 1.67% at 100–150 mm and 0.59% at 150–200 mm. Therefore, the water flow mainly existed at 0–50-mm soil depth in the original grassland, while in the shrub patches and interspace grass patches, water flow was distributed throughout the 0–200-mm soil layer.

3.4 | The relationships between soil macropores, roots and water flow parameters

Correlations between the soil macropore parameters, root parameters and water flow parameters are shown in Table 6. The results show that under the shrub patches the tracer volume density at 70 min was significantly and negatively correlated with root angle. Under the interspace grass patches and the original grassland, the tracer volume density at 35 and 70 min was significantly and positively correlated with the soil macropore node density and root volume density, respectively. Table 7 shows the correlations between the soil macroporosity of different diameters, root volume density of different diameters and tracer volume density at 35 and 70 min. The results showed that the tracer volume density at 35 min was significantly and

TABLE 6 Correlations between the tracer volume density and soil macropore parameters, root parameters under shrub patches, interspace grass patches and original grassland concerning all treatments

Soil macropore parameters		Root parameters												
		Surface						Root						
Treatments	Water flow parameters	Macroporosity density	Length area density	Surface area density	Mean equivalent diameter	Hydraulic radius	Node density	Number density	Root volume density	Mean equivalent diameter	Mean volume	Mean angle	Surface area density	Length density
Shrub patches	The tracer volume density (35 min)	0.274	−0.885	−0.446	−0.362	0.914	−0.650	−0.864	0.560	0.681	0.996	−0.696	0.885	−0.495
	The tracer volume density (70 min)	0.433	−0.794	−0.288	−0.514	0.970	−0.512	−0.766	0.412	0.547	0.968	−0.808 ^a	0.793	−0.635
Interspace grass patches and original grassland	The tracer volume density (35 min)	0.653	0.170	0.677	−0.123	0.414	0.759 ^a	0.139	0.490	0.207	−0.223	0.125	0.548	0.006
	The tracer volume density (70 min)	0.546	0.131	0.552	−0.208	0.406	0.653	0.068	0.779 ^a	0.265	−0.352	0.160	0.693	0.086

Note: Bold indicates correlation is significant at the 0.05 level.
^aCorrelation is significant at the 0.05 level.

TABLE 7 Correlations between water flow and soil macroporosity of different diameters and root volume density of different diameters

Water flow parameters	Equivalent diameter of soil macropores							Equivalent diameter of roots				
	0–1 mm	1–2 mm	2–4 mm	4–6 mm	6–8 mm	8–10 mm	>10 mm	1–2 mm	2–3 mm	3–4 mm	4–5 mm	>5 mm
	0–1 mm	1–2 mm	2–4 mm	4–6 mm	6–8 mm	8–10 mm	>10 mm	1–2 mm	2–3 mm	3–4 mm	4–5 mm	>5 mm
The tracer volume density—35 min	−0.042	0.059	0.036	−0.038	−0.058	0.680^a	0.188	0.806	0.405	−0.137	0.664	−0.102
The tracer volume density—70 min	−0.037	0.072	0.006	−0.086	−0.316	0.569	0.388	0.838^a	0.547	−0.008	0.413	0.029

Note: Bold indicates correlation is significant at the 0.05 level.
^aCorrelation is significant at the 0.05 level.

positively correlated with soil macropores of diameters between 8 and 10 mm. In addition, the tracer volume density at 70 min was significantly correlated with a

TABLE 8 The relative contributions of non-root soil macropore and root to the water flow in all soil columns

The dyed time	Soil macropore/ root	The relative contribution (%)
The dyed volume density— 35 min	Soil macropore	58.84
	Root channel	41.16
The dyed volume density— 70 min	Soil macropore	46.55
	Root channel	53.45

root diameter of 1–2 mm. So, roots are potential channels in the future, while pores are actual voids where water and solute can flow.

Figure 8 shows 3-D visualisations of soil macropores, roots and water flow with a tracer time of 35 min in the soil columns. The results clearly show that water was stored mainly in soil macropores at the soil surface layer (0–10 cm), while water moved along roots at the middle layer (10–30 cm). It is interesting to note that, when the dyed time was 35 min, water flow through soil macropores was greater than along roots and contributed 41.16% to the total water flow (Table 8). However, when the dyed time was 70 min, water flow along roots contributed as high as 53.45% to the total water flow.

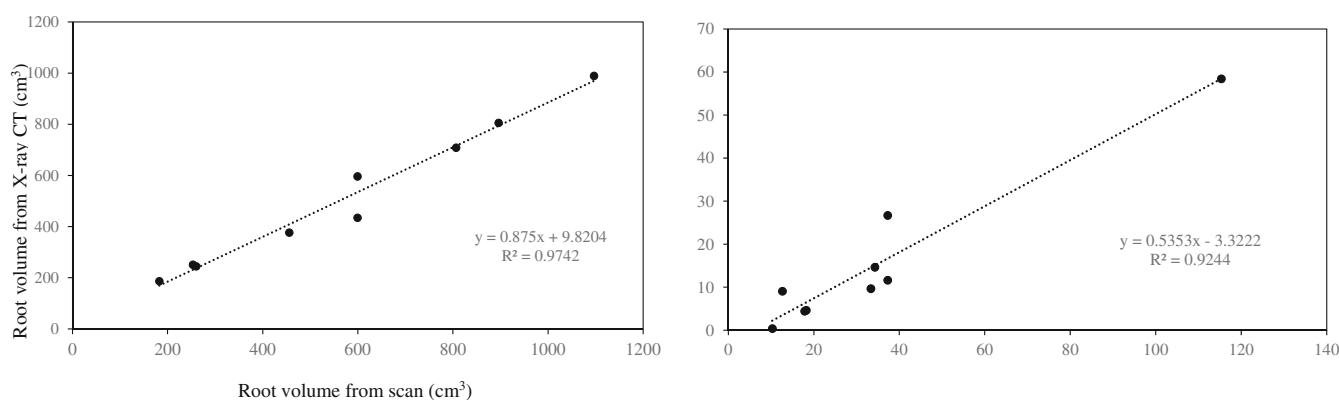


FIGURE 4 Comparisons between the root length and root volume determined by X-ray CT and scanning after washing

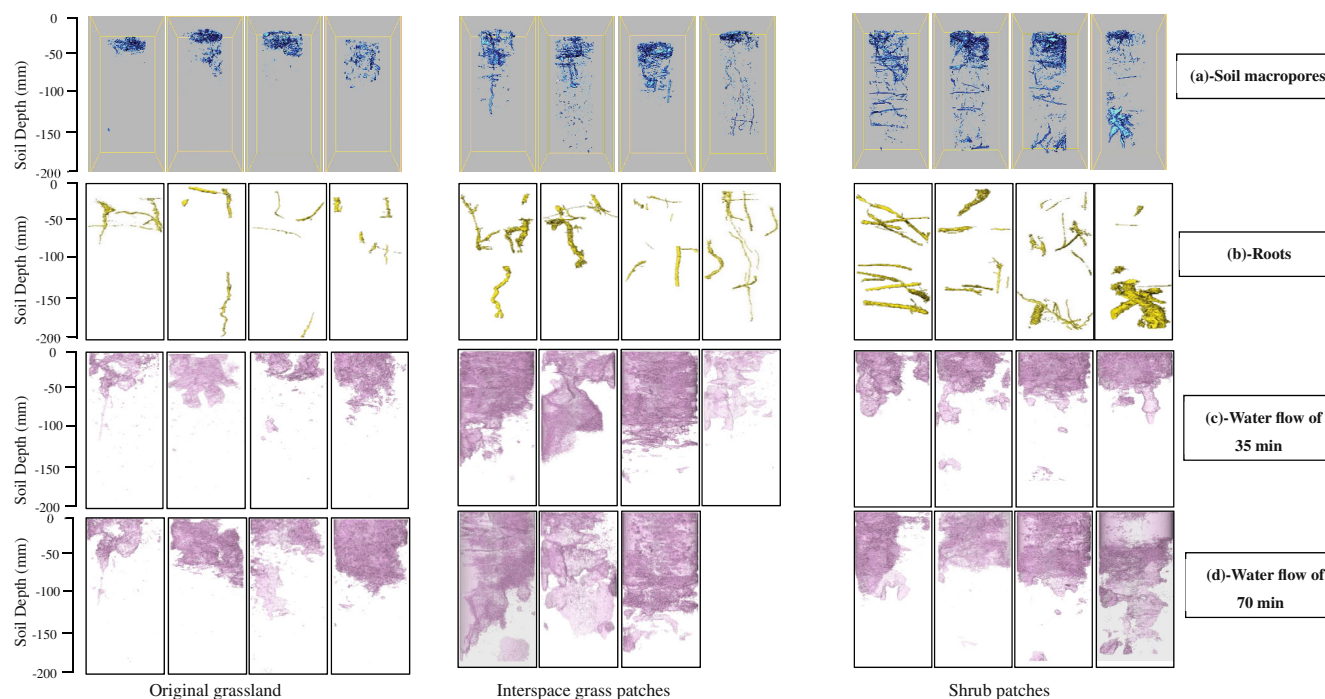


FIGURE 5 3-D visualisations of soil macropores (a) roots (b) water flow networks of 35 min (c) and 70 min (d) in the soil columns

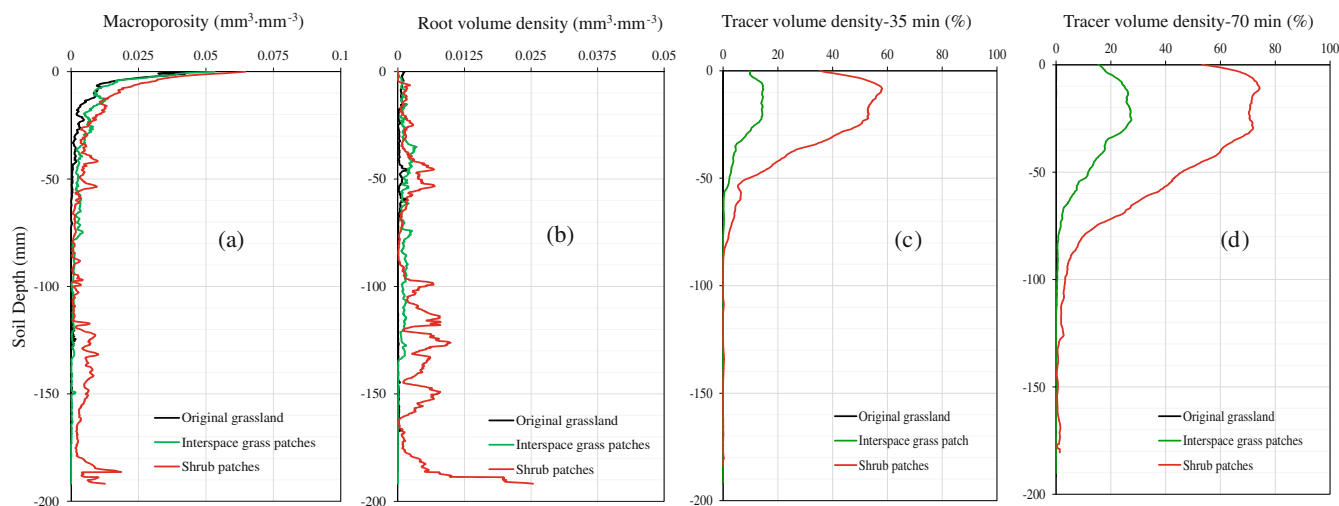


FIGURE 6 Distributions of soil macroporosity, root volume density and tracer volume densities (35 min and 70 min) along soil column depths

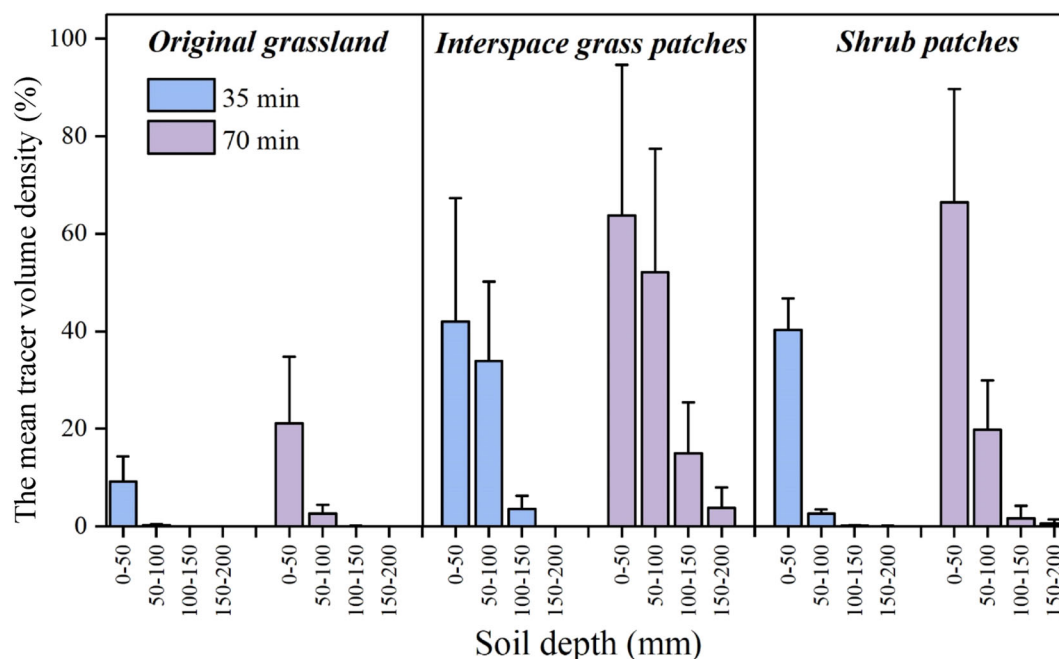


FIGURE 7 The tracer volume density at different soil depths

4 | DISCUSSION

The findings of the study first revealed that the shrub-encroached grasslands (shrub patches and interspace grass patches) had deeper tracer depth than the original grasslands using CT technique (Figure 5c,d and Figure 6c,d), suggesting the shrub encroachment enhanced water flow as compared with non-shrub grassland. Moreover, the tracer volume density was significantly correlated with soil macropore node density, root volume density and root angle (Table 6), suggesting that

shrub roots have a great effect on water flow. These results highlight the direct linkage between water flow and shrub roots and confirmed the hypothesis that water flow varies significantly between shrub-encroached grassland and original grassland and that water flow induced by shrub roots moves soil water into deep soil layers. Previous studies have shown that the root systems of plants created water flow paths by establishing connective preferential pathways (Beven & Germann, 1982; Sarkar et al., 2008). Plant roots improved soil porosity and created macropore flow paths (Feeney et al., 2006; Li

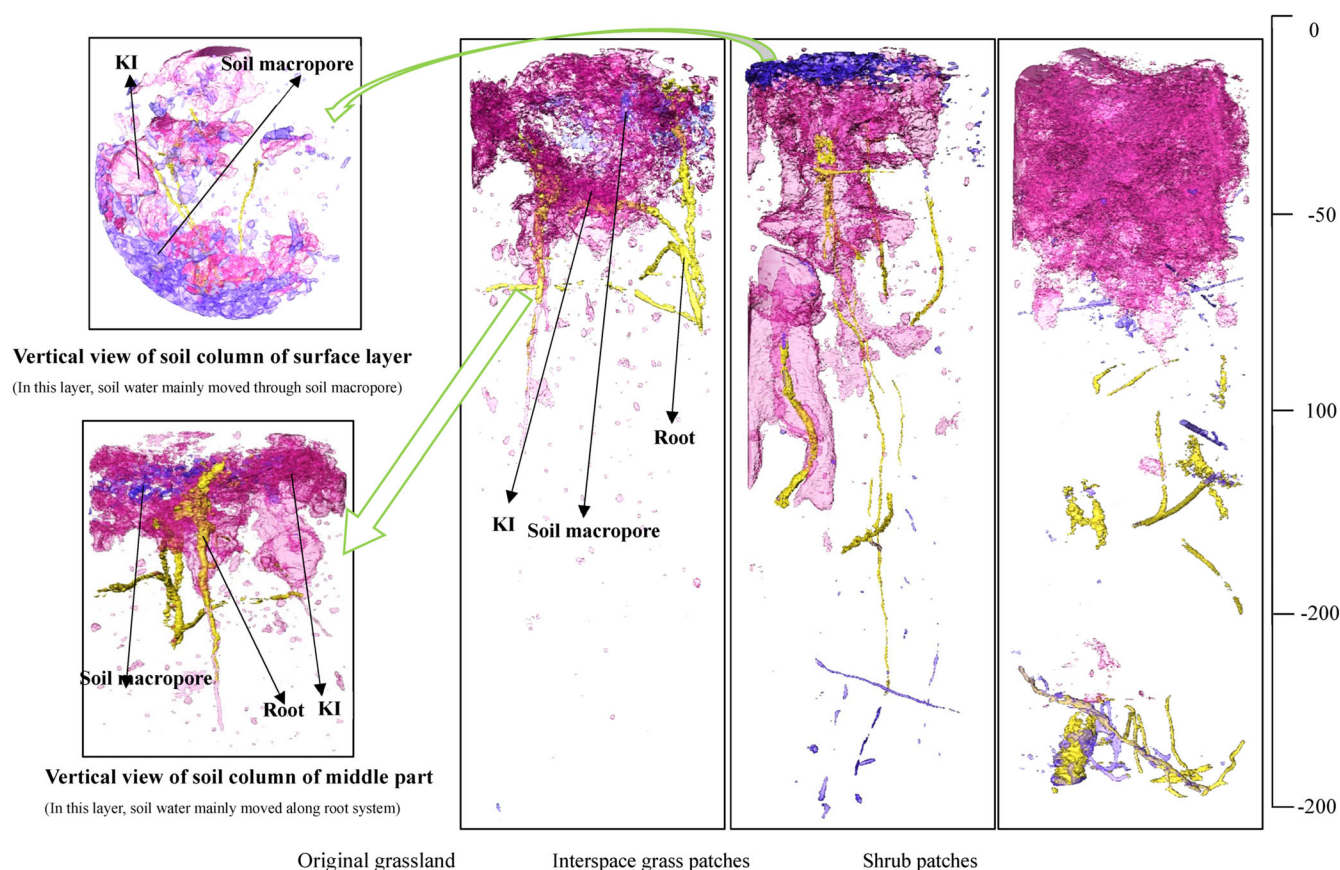


FIGURE 8 Three-dimensional visualisation of root, soil macropore and water flow networks with tracer times of 35 min in the soil columns. Purplish red denotes KI, blue denotes soil macropore and yellow represents roots

et al., 2009). Shougrakpam et al. (2010) revealed that the effective soil macropores for a preferential flow could be attributed to root systems. Chisci et al. (2001) have shown that plant root network in the “Ah horizon” favoured soil macropore and macropore flow. Therefore, soils under the shrub patches and interspace grass patches had deeper tracer depth than that under the original grassland, which was attributed to soil macropore node density, root volume density and root angle.

To explore which part of the root diameter has a significant impact on water flow, we conducted a correlation analysis between the tracer depth and different root diameters (Table 7). There was a significant positive correlation between the tracer depth at 70 min and root with diameters of 1–2 mm. Some previous studies showed that large amounts of well-connected fine roots can form complex networks and enhance the rapid movement of water and solute (Singh et al., 2015), and the presence of fine roots is closely associated with the increase of non-capillary porosity and infiltration capacity (Jiang et al., 2018). Generally, fine roots and the rhizosphere exudates associated with microorganisms can increase organic matter, which subsequently supports soil

aggregation and the formation of inter-aggregate pores (Fageria & Stone, 2006; Singh et al., 2015). Zhang et al. (2015) indicated that higher fine root length density and larger biomass are usually found in preferential flow paths as compared to the soil matrix in field experiments. Hence, shrub fine roots influenced water flow considerably at 70 min time.

The study also showed that the tracer depth of the interspace grass was larger than that of the shrub patches (Figures 5 and 6). This may be due to the root angle. The roots under the shrub patches had a larger mean angle (54.932°) than that of the interspace grass (28.545°), indicating that roots were predominantly oriented horizontally under the shrub patches, through which water could not move to the deeper soil layer. This is consistent with the results of a previous field study, which showed that soil water contents under the interspace grass patches were higher than those under the shrub patches at a soil depth of 10 and 20 cm (Li, Hu, et al., 2013; Li, Zhang, et al., 2013). Therefore, the interspace grass patches had deeper tracer depth than the shrub patches, which was attributed to the vertical-oriented roots under interspace grass patches and horizontal-oriented roots under the shrub patches.

This study highlights the effectiveness of X-ray CT in combination with KI tracer for quantifying water flow, which provides direct evidence of the relations between water flow, soil macropores and especially roots in the shrub-encroached grassland. Our samples were 30 cm in height and 10 cm in diameter to study water flow at the pedon scale. However, due to the trade-off between the sample size and scanning resolution (Allaire et al., 2009; Lucas et al., 2019a; Lucas et al., 2019b), some fine roots were missing and the root networks were not as continuous as expected (Figure 4b). Therefore, future studies need to have a higher spatial resolution and use algorithm methods simultaneously to interpret roots more precisely, which is crucial for the non-destructive quantification of plant roots (Gao et al., 2019). Due to the trade-off between the sample size and scanning speed, the resulting image in the study was a combination of a certain space and time evolution of the tracer. Therefore, future studies need to improve the speed and use suitable algorithms to improve the precision of estimating water flow considering different soil water conditions.

5 | CONCLUSIONS

The results showed that water flow path and spatial distribution could be identified through the combination of KI tracer and CT scanning in undisturbed soils. Soil water moved deeper in the shrub-encroached grassland than in the original grassland. The tracer depth was significantly positively correlated with root equivalent diameter ($p < 0.05$), and the water flow was attributed to the fine roots in the soil. The shrub patches had shallower tracer depth than the interspace grass patches, which was attributed to the horizontal-oriented roots under shrub patches. The relative contribution of water flow along roots was larger than through non-root soil macropores with the increase of dyed time. This provided direct evidence that shrub roots were the main pathway of water flow resulting in water into the deep soil layer in the shrub-encroached grassland. Therefore, shrub-encroached grassland could facilitate greater water flow to deeper soil layers.

AUTHOR CONTRIBUTIONS

Xia Hu: Conceptualization (equal); visualization (equal); writing – original draft (equal). **Xiao-yan Li:** Funding acquisition (equal); supervision (equal); writing – review and editing (equal). **Yun-duo Zhao:** Writing – review and editing (equal). **Ya-Qian Cheng:** Data curation (equal); methodology (equal); supervision (equal). **Zhou Gao:** Data curation (equal). **Zhi-guang Yang:** Data curation (equal).

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DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article

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